

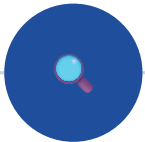
NETRAX

AI-Powered Email Threat Detection, GeoLocation & Forensic Intelligence Platform

“Others detect the threat. NetraX investigates the threat.”



Suspicious
Email



AI
Investigation



Forensic
Evidence



Infrastructure
Intelligence



Risk /
Action

PROBLEM STATEMENT ID	THEME	PS CATEGORY
SIH26106	Blockchain & Cybersecurity	Software
TEAM ID	TEAM NAME	DEPARTMENT
[TEAM ID]	Nexora	Cyber Security Cell
PROBLEM STATEMENT TITLE		
AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform		

All India Council for Technical Education (Cyber Security Cell)

Decision-support & forensic-intelligence prototype — not a substitute for banks, police or official cybercrime reporting (cybercrime.gov.in).



❖ Proposed Solution

NetraX is an agentic-AI platform that investigates a suspicious email beyond simple phishing detection — turning it into cited, correlated forensic evidence.

❖ How It Addresses the Problem

It analyzes email content, headers, sender identity, URLs/domains, authentication signals, and source-IP infrastructure — then correlates that evidence to support forensic investigation, instead of returning a bare suspicious/not-suspicious label.

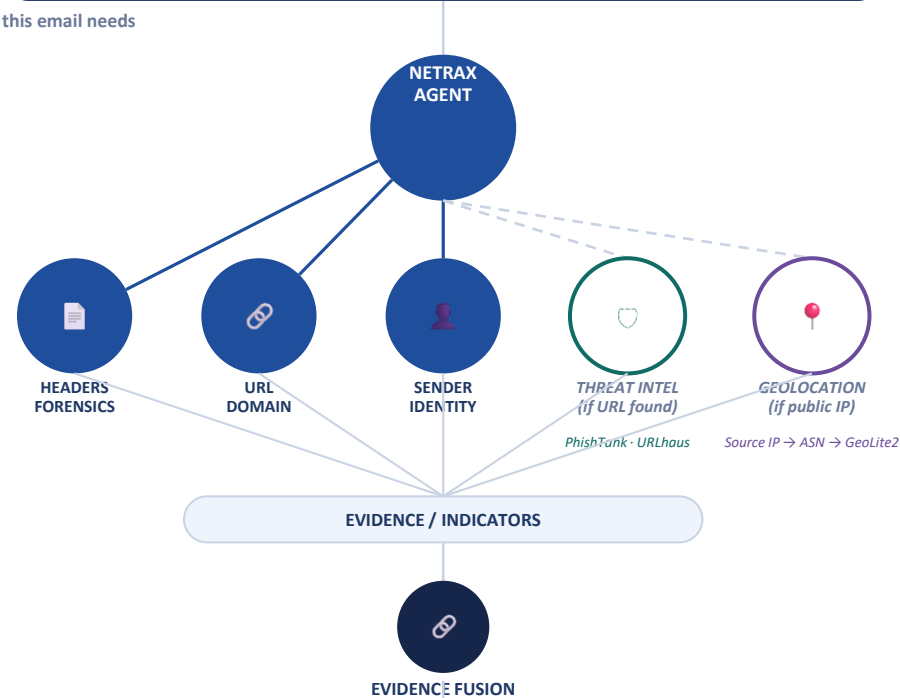
❖ Innovation & Uniqueness

- **Agentic tool selection** — the agent dynamically chooses which investigation tools a case needs; nothing runs blindly.
- **Email forensics** — goes beyond message-content classification into headers, SPF/DKIM/DMARC and Received-chain analysis.
- **Threat intelligence** — adds external evidence via PhishTank and URLhaus correlation.
- **Infrastructure context** — IP → ASN → approximate geolocation, restricted to public source IPs only.
- **Evidence Graph** — connects Email → Sender → Domain → URL → IP → ASN → Geo → Threat Intel.
- **Explainable risk** — a deterministic, per-source-capped score that supports — not replaces — investigator judgment.

01 INPUT



02 AGENT — decides which tools this email needs



04 DECIDE



FRONTEND	BACKEND	ML	AI / TI / GEO
React · TypeScript · Vite · Tailwind CSS · shadcn/ui	Supabase · PostgreSQL · Edge Functions	Python · Pandas · NumPy · scikit-learn	Gemini API (explanation, planned) · PhishTank · URLhaus · MaxMind GeoLite2

Datasets

- Enron Email Corpus — language/structure reference only (not a malicious/benign label).
- SpamAssassin Corpus — spam/ham email corpus; trains the email classifier.
- UCI Phishing Websites — trains the URL classifier.

Models — kept separate

- Email/text — TF-IDF + Linear SVM. F1=0.961, ROC-AUC=0.998 (measured, held-out SpamAssassin set).
- URL/structured — HistGradientBoosting. F1=0.954 (measured, UCI Phishing set).

Two independent, specialized models — not one combined model. The agentic layer orchestrates tool selection; the risk engine performs deterministic scoring.

Threat Intel & Geo

- PhishTank / URLhaus — phishing / malware-URL threat intelligence.
- MaxMind GeoLite2 — IP → ASN → geolocation enrichment.

❖ Why Is NetraX Feasible?

- Public research email and phishing datasets are available for model development and validation.
- Modular Python ML inference integrates with the investigation platform as a separate service.
- Threat intelligence and IP/ASN/geolocation are integrated through dedicated, testable adapters.
- Supabase/PostgreSQL stores cases, evidence, indicators and investigation events at scale.

Challenges

- Data quality
- False positives / false negatives
- Unknown / new attack patterns
- Threat intelligence / API availability
- Security / malicious input
- Scalability

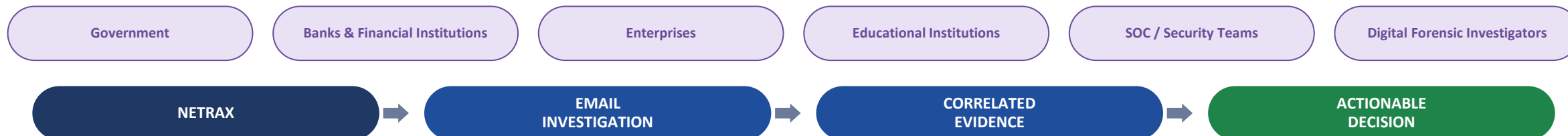
❖ Deployment Feasibility

Mitigation

- Preprocessing + validation before training (reports/data_quality_report.md)
- Model evaluation + decision-threshold tuning on held-out data
- Multi-signal investigation (headers + URL + ML + threat intel) + feedback loop
- Adapter architecture + cached/demo data; reports “unavailable,” never fabricated
- Input validation + HTML sanitization + SSRF protection + secret management
- Modular services + asynchronous investigation steps



❖ Target Audience



Operational Benefits

- Faster suspicious-email triage
- Centralized investigation evidence
- Reduced manual correlation effort
- Explainable risk assessment

Forensic Benefits

- Header and sender analysis
- URL / domain relationships
- IP / ASN infrastructure context
- Evidence Graph for investigation

Security Benefits

- Earlier identification of phishing indicators
- Threat-intelligence correlation
- Better incident investigation
- Structured case reporting

❖ Before vs. After NetraX

Before: Suspicious Email → Manual Checks → Scattered Evidence → Slow Investigation

After: Suspicious Email → NetraX Agentic Investigation → Correlated Evidence → Forensic Risk → Case Report

Datasets

- **CMU Enron Email Dataset**
cs.cmu.edu/~enron
- **Apache SpamAssassin Public Corpus**
spamassassin.apache.org
- **UCI Phishing Websites Dataset**
archive.ics.uci.edu/dataset/327

Threat Intelligence

- **PhishTank**
phishtank.org
- **URLhaus (abuse.ch)**
urlhaus.abuse.ch

Geolocation

- **MaxMind GeoLite2**
dev.maxmind.com/geoip/geolite2-free-geolocation-data

❖ Research Areas

Email phishing detection

Email / header forensics

URL phishing detection

Threat intelligence correlation

Agentic / tool-based investigation

Explainable cybersecurity

Only verified, currently-used sources are listed above — no fabricated papers, DOIs or citations.